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REVIEW PAPER

THE PARAMETERIZATION OF ORTHOGONAL MATRICES:

A REVIEW MAINLY FOR STATISTICIANS

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Summary: Some techniques and statistical applications for the parameterization of orthogonal matrices are surveyed.

1. INTRODUCTION

Some methods of parameterizing orthogonal matrices have appeared in articles and books written by statisticians (see, for example, Anderson, Olkin and Underhill, 1987; Chang, 1986; and Muirhead, 1982). Our aim is to make these methods more widely known to the statistical community by providing the first review of this topic. We do not claim mathematical originality, although probably a few of our comments have some novelty. We confine our review of applications to the statistical ones.

By an orthogonal matrix C we mean a *real* square matrix of order $n \times n$ for which CC' is the identity matrix, where C' denotes the transpose of

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C and we require that $\det(C) = 1$ and not -1 . Orthogonal matrices are used frequently in statistics, especially in linear models and multivariate analysis (see, for example, Anderson, 1965; Graybill, 1961, Chap. 11; James, 1954; Muirhead, 1982, Chap. 2; and Searle, 1971, Chap. 2).

The n^2 elements of an orthogonal matrix are subject to $\frac{1}{2}n(n+1)$ constraints. They can therefore be represented by $n^2 - \frac{1}{2}n(n+1) = \frac{1}{2}n(n-1)$ independent parameters. A representation is convenient if the entire matrix can be computed quickly from the $\frac{1}{2}n(n-1)$ parameters. Such a representation facilitates the search for an orthogonal matrix that satisfies some optimality criterion. The independent parameters can be used also in integrating a function over the n -dimensional orthogonal (or rotation) group (Murnaghan, 1938, pp. 230-242), which is defined as the set of all $n \times n$ orthogonal matrices with the operation of matrix multiplication and is denoted by $O(n)$.

Orthogonal matrices of determinant 1 represent elements of the rotation group in n dimensions. Gel'fand, Minlos, and Shapiro (1963) and Murnaghan (1938) give extensive discussions of the representation of the n -dimensional rotation group. Hoffman, Raffinetti, and Ruedenberg (1972) and Raffinetti and Ruedenberg (1970) represented the class of n -dimensional orthogonal matrices in terms of generalized Eulerian angles. FORTRAN programs were written by these authors to construct an orthogonal matrix of order $n \times n$ from $\frac{1}{2}n(n-1)$ Eulerian angles, and vice versa, to determine these angles from a given orthogonal matrix. The representation of three-dimensional orthogonal matrices by Eulerian angles has long been used in the study of the motion of a rigid body (Euler, 1776; Whittaker, 1927, pp. 9-10). For further ideas and a three-dimensional application, see Moran (1975).

2. SOME APPLICATIONS IN STATISTICS OF THE PARAMETERIZATION OF THE ORTHOGONAL GROUP

2.1 THE HAAR MEASURE

A parameterization of $O(n)$ can be used to define an invariant (Haar) measure, μ , on $O(n)$. This measure is invariant in the sense that if a set G of orthogonal $n \times n$ matrices has measure $\mu(G)$, then, for every orthogonal $n \times n$ matrix C , we have $\mu(G) = \mu(GC) = \mu(CG)$, where GC [and CG] denotes the set of matrices obtained by multiplying all the matrices in G on the right [left] by C . Haar measure is unique up to a positive multiplicative constant. In accordance, for example, with James (1954, p. 53), the Haar measure $\mu(G)$, if it exists, is given by the $\frac{1}{2}n(n-1)$ -fold integral

$$\mu(G) = \int_G \prod_{\substack{1 \leq i < j \leq n \\ ij}} (c'_i \cdot dc_j), \quad G \subset O(n)$$

where c_i and c_j are the i -th and j -th column vectors of an orthogonal matrix C , $(c'_i \cdot dc_j)$ denotes a scalar product, and the product of differential forms is to be interpreted as an *exterior product*. (Exterior products are explained, for example, by James, 1954, p. 46.) For example, if $n = 2$, the differential form, $\Pi(c'_i \cdot dc_j)$, for

$$C = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}, \quad 0 \leq \theta < 2\pi$$

is $(c'_1 \cdot dc_2) = d\theta$.

The Haar measure has been used to derive the distribution of the canonical correlation coefficients (James, 1954). Anderson (1965) used the representation given by formula (3.2) of Section 3 below to obtain the joint distribution of the eigenvalues of the sample variance-covariance matrix. See also, for example, Wijsman (1957), Cattopadhyay, Pillai, and Li (1976).

2.2 GENERATION OF RANDOM ORTHOGONAL MATRICES

When carrying out simulation experiments in regression theory, the condition number of $\mathbf{X}'\mathbf{X}$ is of interest, where $\mathbf{X}$ is the matrix associated with the regression model. There are several definitions for the condition number, one of which is the ratio of the largest to the smallest of the eigenvalues of $\mathbf{X}'\mathbf{X}$ (see, for example, Turing, 1948, and Hartree, 1955, p. 153). (For a brief introduction to the topic see Stewart, 1985.) It seems sensible to "stratify" the simulation experiments according to condition numbers, that is, to generate several different matrices $\mathbf{X}$ for which the condition numbers of $\mathbf{X}'\mathbf{X}$ are equal. This can be done (as George Terrell pointed out in conversation) by applying random orthogonal transformations to $\mathbf{X}$. Generation of random orthogonal matrices was used by Heiberger, Velleman, and Ypelaar (1983) to construct test data with special properties for multivariate linear models. Such data can be employed to evaluate the performance characteristics of statistical methods and programs. The algorithm described by these authors permits the user to control, among other things, the structure of the $\mathbf{X}$ matrix. The task of generating random orthogonal matrices can be facilitated by parameterization. This was also proposed by Anderson, Olkin and Underhill (1987).

It was shown succinctly by Dawid (1977, p. 257) that random orthogonal matrices, in the Haar-measure sense, can be generated by using the following elegant procedure (mentioned also by Anderson, Olkin and Underhill, 1987). First generate an $n \times n$ matrix $\mathbf{Z} = (z_{ij})$ where the z_{ij} are independent standard normal variables. Then decompose $\mathbf{Z}$ into its singular decomposition,

$$\mathbf{Z} = \mathbf{W}_1 \mathbf{A} \mathbf{W}_2'.$$

Then $\mathbf{W}_1$ and $\mathbf{W}_2$ are independent $n \times n$ orthogonal matrices both random in the Haar-measure sense. We thus get two random orthogonal matrices for each selection of $\mathbf{Z}$.

A possible disadvantage of this procedure, when the "Haar randomness" of the orthogonal matrices is not required, is that the procedure involves

n^2 parameters, whereas the dimensionality of the space of orthogonal matrices is only $n(n-1)/2$. In a search for an "optimal" orthogonal matrix, whatever the definition of optimality, the excess of parameters might be awkward.

Other methods for generating random orthogonal matrices were given by Heiberger (1978) (with a remark by Tanner and Thisted, 1982), Stewart (1980), and Anderson, Olkin and Underhill (1987). Stewart's method is based on expressing an orthogonal matrix as the product of Householder transformations. A Householder transformation is a symmetric orthogonal matrix of the form

$$H = I - \frac{2uu'}{u'u}$$

where u is a nonzero vector. The matrix belongs to a more general class of elementary orthogonal matrices whose properties were studied by Constantine and Gower (1978a). Stewart used his method to describe briefly two techniques for estimating the condition number of a general matrix.

2.3 DIAGONALIZATION OF SEVERAL MATRICES

The simultaneous diagonalization of several matrices (see, for example, Graybill, 1983, p. 406) is of considerable interest in statistics, especially in the analysis of random- or mixed-effects models. Hultquist and Atzinger (1972), for example, were able to obtain minimal sufficient statistics for the parameters in a general mixed-effects model through a nonsingular transformation of the data vector y . This transformation reduces the variance-covariance matrix, Σ , of y to a diagonal matrix provided that the design matrices that lead to Σ are simultaneously diagonalizable.

Let $A_1, A_2, \dots, A_\ell$ be $\ell (\geq 2)$ positive definite symmetric matrices of order $p \times p$. Suppose that these matrices are not simultaneously diagonalizable (because they do not commute), and we wish to find an orthogonal matrix, B , which makes them simultaneously "as diagonal as possible". Flury and Gautschi (1986, p. 170) introduced a measure of simultaneous deviation of the matrices, $BA_1B', BA_2B', \dots, BA_\ell B'$, from diagonality. This

measure is given by the function

$$(2.1) \quad \chi(\mathbf{B}, \mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_\ell; n_1, n_2, \dots, n_\ell) \\ = \prod_{i=1}^{\ell} \{ |\text{diag}(\mathbf{B}\mathbf{A}_i\mathbf{B}')| / |\mathbf{B}\mathbf{A}_i\mathbf{B}'| \}^{n_i},$$

where $\text{diag}(\mathbf{B}\mathbf{A}_i\mathbf{B}')$ is a diagonal matrix having the same diagonal elements as $\mathbf{B}\mathbf{A}_i\mathbf{B}'$ ($i = 1, 2, \dots, \ell$), $|\cdot|$ denotes the determinant of a matrix, and $n_1, n_2, \dots, n_\ell$ are positive weights. It is known that $\chi \geq 1$ with equality occurring if and only if $\mathbf{B}$ diagonalizes the $\mathbf{A}_i$'s. It is therefore of interest to find the minimum of χ over the rotation group $O(p)$. For an application see Flury (1984) and Flury and Gautschi (1986).

The determination of the optimal orthogonal matrix, $\mathbf{B}_0$, that minimizes χ in (2.1) can be reduced to an optimization problem in $\frac{1}{2}p(p-1)$ dimensions since this is the number of independent parameters that represent the elements of $O(p)$. Flury and Gautschi (1986) did not follow this procedure; instead, they introduced an iterative algorithm whereby a converging sequence of orthogonal matrices, $\mathbf{B}^{(0)}, \mathbf{B}^{(1)}, \dots$, was derived such that $\chi(\mathbf{B}^{(j+1)}) \leq \chi(\mathbf{B}^{(j)})$, $j = 0, 1, 2, \dots$. All pairs of columns of the orthogonal matrix $\mathbf{B}^{(j)}$ obtained in the j^{th} iteration ($j = 0, 1, 2, \dots$) are subjected to rotations by a specific 2×2 orthogonal matrix parameterized by a single parameter. This yields the matrix $\mathbf{B}^{(j+1)}$ and the process is repeated until some convergence criterion is met.

2.4 DESIGN-RELATED ROBUSTNESS

The need for parameterization arises in response surface analysis. Consider a linear model of the form

$$(2.2) \quad \mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

where $\mathbf{y}$ is the vector of observations, $\boldsymbol{\beta}$ is a vector of unknown parameters, and $\mathbf{X}$ is a known matrix of order $n \times p$ and rank p . The elements of $\mathbf{X}$ are

functions of the settings of m input variables denoted by $x_1, x_2, \dots, x_m$. The $n \times m$ matrix, $D = \{x_{ui}\}$, where x_{ui} is the setting of the i^{th} variable at the u^{th} experimental run ($u = 1, 2, \dots, n; i = 1, 2, \dots, m$), is called the design matrix. The elements of the error vector ϵ in (2.2) are assumed to be independently and identically distributed random variables with zero means and variances equal to σ^2 .

Tests of significance concerning the parameter vector β in (2.2) depend on the assumption that ϵ is normally distributed. The effect of nonnormality of the error distribution on these tests has been studied by several authors (see, for example, Pearson, 1931; Geary, 1947; Gayen, 1950; and David and Johnson, 1951a, 1951b). Box and Watson (1962) pointed out that the sensitivity to nonnormality depends very much on the settings of the input variables specified in the design matrix. This was also demonstrated by Vuchkov and Solakov (1980). Thus, a properly chosen design matrix can lead to tests that are resistant, or robust, to failure of the normality assumption. Khuri and Myers (1981) presented an algorithm for the construction of designs that satisfy the robustness criterion introduced by Box and Watson (1962). Parameterization of orthogonal matrices played a central role in the execution of this algorithm. For this application it is not necessary for the orthogonal matrices to be generated in accordance with Haar measure.

We now describe various methods of parameterization.

3. HOW TO PARAMETERIZE AN ORTHOGONAL MATRIX

The four methods we shall discuss are (i) expressing an orthogonal matrix in the form $e^{\mathbf{T}}$ where $\mathbf{T}$ is skew-symmetric, (ii) using Cayley's transformation which also gives an expression in terms of a skew-symmetric matrix, (iii) using Eulerian angles for the three-dimensional case, and (iv) using generalized Eulerian angles for the n -dimensional case. We now explain these four methods.

3.1 EXPONENTIAL TRANSFORMATION

If Q is an $n \times n$ orthogonal matrix with determinant 1, then it can be written in the form

$$(3.1) \quad Q = e^T,$$

where T is a skew-symmetric matrix (see, for example, Gantmacher, 1959, p. 288). The elements of T , above its main diagonal, can be used to parameterize Q . The exponential mapping (3.1) is defined by the infinite series

$$e^T = I + T + \frac{T^2}{2!} + \frac{T^3}{3!} + \dots$$

The exponential parameterization was used in a theorem concerning the asymptotic joint density function of the eigenvalues of the sample covariance matrix (Muirhead, 1982, p. 394) and in a study concerning asymptotic tests and confidence regions for an unknown orthogonal matrix of rotation of a sphere of radius 1 in an-dimensional Euclidean space (Chang, 1986).

If Q is given, to find T we first find the eigenvalues of Q . These are of the form $e^{\pm i\phi_1}, e^{\pm i\phi_2}, \dots, e^{\pm i\phi_q}, 1$, where the eigenvalue 1 is of multiplicity $(n - 2q)$, and one of the real number ϕ_j is a multiple of 2π ($j = 1, 2, \dots, q$). (Those that are odd multiples of π give an even number of eigenvalues equal to -1 .) If we denote the matrix

$$\begin{bmatrix} a & b \\ -b & a \end{bmatrix}$$

by $[a + bi]$, a notation that is reasonable because the 2×2 matrices form a representation of complex numbers, then Q can be written (Gantmacher, 1959, p. 288), as the product of three *real* matrices,

$$(3.2) \quad Q = Q_1 \text{diag}\{[e^{i\phi_1}], \dots, [e^{i\phi_q}], 1, \dots, 1\} Q_1'$$

Here Q_1 is an orthogonal matrix of the form

$$Q_1 = [x_1, y_1, x_2, y_2, \dots, x_q, y_q, x_{2q+1}, \dots, x_n]$$

such that $x_j + iy_j$ is an eigenvector with eigenvalue $e^{i\phi_j}$ ($j = 1, \dots, q$), and x_k is an eigenvector with eigenvalue 1 ($k = 2q + 1, \dots, n$). We now define the skew-symmetric matrix T by the equation

$$(3.3) \quad T = Q_1 \text{diag}\{[i\phi_1], \dots, [i\phi_q], 0, \dots, 0\} Q_1'.$$

Since $e^{[i\phi_j]} = [e^{i\phi_j}]$ we have $Q = e^T$ as required. We note that

$$Q = \prod_{j=0}^{q-1} R_j$$

where

$$R_j = Q_1 \text{diag}\{I_{2j}, [e^{i\phi_{j+1}}], I_{n-2j-2}\} Q_1', j = 0, \dots, q-1,$$

and where I_{2j} and I_{n-2j-2} are the identity matrices of dimensions $2j$ and $n - 2j - 2$, respectively. If Π_{j+1} is a plane parallel to both the vectors x_{j+1} and y_{j+1} , then R_j represents a rotation of the n -space leaving fixed every point in the $(n - 2)$ -space orthogonal to Π_{j+1} (Vitali, 1928). Thus Q represents the product of q such simple rotations. When $n = 3$ and $q = 1$, this is a familiar fact in dynamics, known to Euler. To obtain a power Q^α we could multiply all the angles of rotation by α . We note also that

$$(3.4) \quad \begin{aligned} T(x_j + iy_j) &= i\phi_j(x_j + iy_j), \quad j = 1, 2, \dots, q, \\ T x_k &= 0, \quad k = 2q + 1, \dots, n. \end{aligned}$$

Equations (3.4) state that $x_j + iy_j$ is an eigenvector of T with eigenvalue $i\phi_j$, and that x_k is an eigenvector of T with zero eigenvalue. Since the eigenvalues of T are purely imaginary or zero (at least one of them is zero if n is odd), and the imaginary ones pair off in conjugate pairs $\pm i\phi_1, \dots, \pm i\phi_q$, it follows that (3.3) gives a representation of a real skew-symmetric matrix in terms of its eigenvalues and eigenvectors (Gantmacher, 1959, p. 285). This observation enables us to calculate Q if

$\mathbf{T}$ is given and if (3.1) is known to be true. We first put $\mathbf{T}$ in the form (3.3), then we calculate $\mathbf{Q}$ by (3.2). For convenience in computing the eigenvalues and eigenvectors of $\mathbf{T}$, we note that $\mathbf{x}_j$ and $\mathbf{y}_j$ ($j = 1, \dots, q$) are eigenvectors of the symmetric matrix $\mathbf{T}^2$ with eigenvalue $-\phi_j^2$. This can be easily seen from (3.3), which when both sides are squared gives

$$\mathbf{T}^2 = - \sum_{j=1}^q \phi_j^2 (\mathbf{x}_j \mathbf{x}_j' + \mathbf{y}_j \mathbf{y}_j').$$

Aside from their close relationship with orthogonal matrices, skew-symmetric matrices have interesting applications in statistics (see Constantine and Gower, 1987b; Gower, 1977). Gower (1980) discussed properties pertaining to the eigenvectors of skew-symmetric matrices. The latter article lists several references to the author's earlier work.

Example, $n = 3$. In three dimensions any orthogonal matrix $\mathbf{Q}$, with determinant 1, has the form

$$\mathbf{Q} = \exp \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}.$$

The eigenvalues of

$$\mathbf{T} = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}.$$

are $i\rho$, $-i\rho$, and 0, where $\rho = (a^2 + b^2 + c^2)^{\frac{1}{2}}$. The normalized eigenvector with eigenvalue 0 is $\mathbf{w} = \rho^{-1}[a, b, c]'$ and the other normalized eigenvectors can be chosen as $\mathbf{u} \pm i\mathbf{v}$ where

$$\mathbf{u} = (b^2 + c^2)^{-\frac{1}{2}}[0, c, -b]'$$

and

$$\mathbf{v} = \rho^{-1}(b^2 + c^2)^{-\frac{1}{2}}[b^2 + c^2, -ab, -ac]'$$

The vectors $\mathbf{u}$ and $\mathbf{v}$ are orthonormal eigenvectors of $\mathbf{T}^2$ with the eigenvalue $-\rho^2$. From (3.2), we have

$$\mathbf{Q} = [\mathbf{u}, \mathbf{v}, \mathbf{w}] \text{diag}\{[e^{-i\rho}], 1\}[\mathbf{u}, \mathbf{v}, \mathbf{w}]'.$$

The matrix $\mathbf{Q}$ represents a rotation of the 3-space about a line whose direction cosines are $(a/\rho, b/\rho, c/\rho)$.

3.2 CAYLEY'S TRANSFORMATION

If $\mathbf{Q}$ is an orthogonal matrix that does not have the eigenvalue -1 , then it may be written in Cayley's form (Gantmacher, 1959, p. 289),

$$(3.5) \quad \mathbf{Q} = (\mathbf{I} - \mathbf{S})(\mathbf{I} + \mathbf{S})^{-1},$$

where $\mathbf{S}$ is a skew symmetric matrix. [$\mathbf{I} + \mathbf{S}$ is never singular if $\mathbf{S}$ is skew-symmetric. A proof of this fact is given by Ferrar (1950, p. 163) who attributes the fact to Cayley. A simpler proof is that all the eigenvalues of a skew-symmetric matrix $\mathbf{S}$ are purely imaginary or zero ($i\mathbf{S}$ being Hermitian) and therefore -1 cannot be an eigenvalue of $\mathbf{S}$.] The form (3.5) has the advantage of defining $\mathbf{Q}$ as a one-to-one function of $\mathbf{S}$, but it has the disadvantage of being restricted to those orthogonal matrices that do not have the eigenvalue -1 . If this condition is not assumed, $\mathbf{Q}$ can still be written in the form (Ferrar, 1950, p. 164)

$$(3.6) \quad \mathbf{Q} = \mathbf{J}(\mathbf{I} - \mathbf{S})(\mathbf{I} + \mathbf{S})^{-1}$$

where $\mathbf{J}$ is a diagonal matrix in which each element on the diagonal is either 1 or -1 (and where $\mathbf{S}$ is skew-symmetric). This representation can be made unique by insisting that all the plus 1's precede all the minus 1's along the diagonal of $\mathbf{J}$. When this representation is used we must supplement the $\frac{1}{2}n(n-1)$ parameters in $\mathbf{S}$ with an assumption for the number, r , of plus 1's in $\mathbf{J}$.

The Cayley transformation given in (3.5) was discussed by Olkin and Sampson (1972) and, more recently, by Anderson, Olkin, and Underhill (1987). In the latter article, the authors propose an alternative parameterization based on the product of $n(n-1)/2$ orthogonal matrices, each of which represents an angle of rotation. Such parameterizations can be

used to generate random orthogonal matrices distributed according to the Haar measure over the orthogonal group $O(n)$.

3.3 EULERIAN ANGLES IN THREE DIMENSIONS

In three dimensions orthogonal transformations with determinant 1 are of course represented by the rotation of a rigid body free to turn about a point 0. The motion of this body is determined by three independent angles known as Eulerian angles (Euler, 1776; Whittaker, 1927, p. 9; Condon, 1958, p.6). These angles can be introduced in various ways. The following one is used by Condon:

Let OXYZ be a right-handed system of rectangular axes fixed in space. Let Oxyz be rectangular axes fixed relative to the body and moving with it, such that before the displacement the two sets of axes OXYZ and Oxyz are coincident in position. Let $\mathbf{v}_Z$ and $\mathbf{v}_z$ be unit vectors in the positive directions of the Z and z axes, respectively, and let $\mathbf{K} = \mathbf{v}_Z \times \mathbf{v}_z$ be their vector product. Denote the angles ZOz, XOK, KOx by θ, ϕ, ψ , respectively. These are the three Eulerian angles.

By a remark made in Section 3.1 the total movement of the body is equivalent to a rotation. This rotation is represented by the product of three matrices, thus:

$$\mathbf{Q} = \text{diag}\{[e^{i\psi}], 1\} \text{diag}\{1, [e^{i\theta}]\} \text{diag}\{[e^{i\phi}], 1\} =$$

(3.7)

$$\begin{bmatrix} \cos\phi \cos\psi - \sin\phi \cos\theta \sin\psi & \sin\phi \cos\psi + \cos\phi \cos\theta \sin\psi & \sin\theta \sin\psi \\ -\cos\phi \sin\psi - \sin\phi \cos\theta \cos\psi & -\sin\phi \sin\psi + \cos\phi \cos\theta \cos\psi & \sin\theta \cos\psi \\ \sin\phi \sin\theta & -\cos\phi \sin\theta & \cos\theta \end{bmatrix}$$

This then provides a parameterization in three dimensions. The parameterization is unique, if, for example, $0 \leq \theta < 2\pi$, $0 \leq \phi < 2\pi$, and $0 \leq \psi \leq \pi$.

3.4 GENERALIZED EULERIAN ANGLES

A generalization of the concept of the three-dimensional Eulerian angles to n dimensions was given by Raffenetti and Ruedenberg (1970). Formulae and a computer program were derived that expressed an arbitrary orthogonal matrix of order $n \times n$ in terms of $\frac{1}{2}n(n-1)$ angular variables. A general n -dimensional orthogonal matrix, Q , can be constructed by the sequence of recurrence relations

$$\begin{aligned} Q^{(1)} &= \pm 1 \\ q^{(i)} &= \text{diag}\{Q^{(i-1)}, 1\}, \quad i = 2, 3, \dots, n, \\ A^{(i)} &= a_{i-1,i} a_{i-2,i} \cdots a_{1,i}, \quad i = 2, 3, \dots, n, \\ Q^{(i)} &= A^{(i)} q^{(i)}, \quad i = 2, 3, \dots, n, \\ Q &= Q^{(n)} \end{aligned}$$

where $(a_{i,j})(i = 1, 2, \dots, n; j = 1, 2, \dots, n; i < j)$ is an $n \times n$ matrix whose diagonal elements are 1 except for the diagonal elements in the i^{th} and j^{th} columns, which are $\cos \gamma_{ij}$; all off-diagonal elements are 0 except for the one corresponding to the intersection of the i^{th} row and j^{th} column, which is $\sin \gamma_{ij}$, and that on the intersection of the j^{th} row and the i^{th} column, which is $-\sin \gamma_{ij}$. The choice $Q^{(1)} = 1$ yields an orthogonal matrix $Q = \{Q_{ij}\}$ with determinant 1, and the choice $Q^{(1)} = -1$ yields a matrix Q with determinant -1 .

Conversely, if an orthogonal matrix, $U = \{U_{ij}\}$, is given, then the corresponding angular variables are found by minimizing the function $f(\gamma) = \sum_{i,j} [U_{ij} - Q_{ij}(\gamma)]^2$, and where $\gamma = (\gamma_{12}, \gamma_{13}, \dots, \gamma_{n-1,n})'$ ($i, j = 1, 2, \dots, n$), and the elements of Q are found by a prescribed recurrence scheme. Hoffman, Raffenetti, and Ruedenberg (1972), however, give algebraic inversion formulae, and a computer program, for the angular variables in terms of the elements of the orthogonal matrix.

Tumura (1965) used the representation by generalized Eulerian angles to derive the noncentral distribution of the eigenvalues of a Wishart matrix. This provided an elementary development of zonal polynomials (see An-

derson, Olkin, and Underhill, 1987).

3.5 COMPARISON AMONG METHODS OF PARAMETERIZATION

To our knowledge no formal comparisons have been made of the running times of the algorithms for the various methods of parameterization. It is beyond the scope of the present article to evaluate and compare the numerical efficiencies of these algorithms. However, we shall report here a few observations made by other authors about this subject.

Anderson, Olkin, and Underhill (1987) carried out a simulation study to determine the execution times needed to generate $n \times n$ orthogonal matrices ($n = 4, 6, 8, 10, 12$) using their method of successive rotations. The method by Stewart (1980) for generating random orthogonal matrices from the Haar distribution using Householder transformations seems, according to Sloane (1983, p. 100), to be the most efficient of the methods known to him although he doubted whether a comparison of the running times had been for a specific value of n . The running time required to generate an $n \times n$ orthogonal matrix by Stewart's method is proportional to n^2 .

4. A PROBLEM CONCERNING HAAR MEASURE

It would be of value for numerical integration over the orthogonal group if Haar measure could be generated by $\frac{1}{2}n(n-1)$ statistically independent random angles each uniformly distributed. For we could then approximate the integration over the orthogonal group by a discrete sum over points uniformly spaced on circles. In three dimensions the Eulerian angles do not serve this purpose. For suppose that ϕ, θ , and ψ are each uniformly distributed. Then $\cos\theta$ cannot have the same distribution as the other elements on the diagonal of the matrix in (3.7), so that the transformation is not symmetrical with respect to the axes. Thus the distribution cannot be invariant under all rotations.

One can arrive at Haar measure for the three-dimensional rotation group

by choosing an axis of rotation with uniform polar angles in $(0, \pi)$ and $(0, 2\pi)$ and then choosing an angle ω of rotation with uniform distribution on $(0, 2\pi)$. The matrix giving this orthogonal transformation, expressed in terms of ω and the direction cosines, $\cos\lambda$, $\cos\mu$, and $\cos\nu$, of the axis of rotation, is

$$(4.1) \quad \begin{bmatrix} 1 - 2s_1^2h^2 & 2c_1c_2h^2 - c_3s & 2c_1c_3h^2 + c_2s \\ 2c_1c_2h^2 + c_3s & 1 - 2s_2^2h^2 & 2c_2c_3h^2 - c_1s \\ 2c_1c_3h^2 - c_2s & 2c_2c_3h^2 + c_1s & 1 - 2s_3^2h^2 \end{bmatrix}$$

where $s = \sin\omega$, $h = \sin\frac{1}{2}\omega$, $s_1 = \sin\lambda$, $s_2 = \sin\mu$, $s_3 = \sin\nu$, $c_1 = \cos\lambda$, $c_2 = \cos\mu$, $c_3 = \cos\nu$. This result can be readily derived from Whitaker (1927, p. 8). Formula (4.1) provides a matrix representation of the rotation group and it is equipped with Haar measure if λ, μ, ν and ω have uniform distributions in $(0, 2\pi)$, $(0, 2\pi)$, $(0, 2\pi)$, and $(0, \pi)$, respectively. It seems to be difficult to generalize this result to n dimensions.

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